# Willian Vieira PhD

Applied statistician and R developer with a PhD in quantitative ecology. I take statistical methods from concept to R implementation to evaluation, turning them into reproducible analyses scientists can trust.

## DATA SCIENCE & ENGINEER EXPERIENCE

2025 | 2024 (1 yr 9 m)

### Data Analyst

Habitat, Montreal, Canada

Architected end-to-end R/Python statistical and ML pipelines to model large-scale, noisy real-world data for client-facing analytical products | Owned model validation, QA, and monitoring, with automated tests, sanity checks, and provenance so analyses stayed traceable as data evolved | Translated statistical decisions into clear outputs for non-technical stakeholders, including visualizations, warnings, and explanations of model assumptions | Led the transition to a reproducible R/Unix production environment using Docker, CI/CD, package-style code organization, and code review | Designed supporting cloud data infrastructure on Azure for heterogeneous data and feature retrieval

2024 | 2017

### PhD Research

Integrative Ecology Lab, Sherbrooke, Canada

Developed Bayesian hierarchical and nonlinear statistical models in R and Stan to represent heterogeneous biological populations, forecast dynamics, and propagate uncertainty from noisy, biased, incomplete real-world data | Authored open-source R packages (, ) for demographic modelling, with version control, automated tests, documentation, and reusable APIs | Published a technical methods book walking from mathematical concept to R implementation to model evaluation, alongside one peer-reviewed publication and two preprints ( , ) | Ran thousands of simulations on HPC infrastructure () to test model behavior, compare assumptions, and validate inference at scale | Built reproducible R pipelines harmonizing climate, remote sensing, and field-inventory data across North America

2022 | 2020 (part-time)

### Biostatistician

Environment and Climate Change Canada - Quebec, Canada

Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias-correction methods for large-scale ecological monitoring data, designing an approach later adopted by other provinces | Engineered a fully automated and reproducible data pipeline with version-controlled workflows, automated tests, documentation, and stakeholder-ready analytical outputs

## EDUCATION

2024 | 2017

### PhD, Ecology

Université de Sherbrooke - Sherbrooke, Canada

*How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*

2016 | 2015

### Masters 2, Agroecology and Resource Management

Bordeaux Sciences Agro, Bordeaux, France

*Modelling the dispersion of weed species in agricultural landscapes*

2015 | 2010

### BSc in Agronomy

Universidade Federal de Santa Catarina, Florianópolis, Brazil

## CONTACT


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GitHub/WillVieira
Publications


## SKILLS

**Statistics & Modelling**: Multi-level models · ANOVA · PCA & multivariate analysis · Bayesian hierarchical models · Simulation · Uncertainty quantification · Model evaluation

**Programming & Database**: R · Python · Stan · SQL · Bash · DuckDB · Julia

**Data Engineering**: Package development · testthat · renv · R Markdown / Quarto · Shiny · Code review · Documentation · Reproducible pipelines

**DevOps & Reproducibility**: Git · CI/CD · Unix · Docker · HPC · Automated validation · Provenance · Make

## OUTREACH

· Developed and maintained the QCBS R Workshop Series, teaching R and statistics to nearly one thousand graduate students · Taught statistics, probability, and scientific programming to 250+ students across biology, engineering, and graduate programs · Published 5 papers & preprints with ~100 citations · Translated statistical methods into clear explanations and reproducible analyses for non-specialist audiences

## LANGUAGES

Portuguese · Native
French · Full Professional
English · Full Professional

## TEACHING EXPERIENCE

2021

**Biodiversity Modelling Summer School**

Bios$^2$ summer school — Sherbrooke, CA

Teaching assistant and lectured for graduate students Evaluating ecological models using Bayesian statistics

2020 | 2019

**Methods in computational ecology**

BIO500 - Département de biologie, Université de Sherbrooke — Sherbrooke, CA

Teaching assistant and lectured for undergraduate students Tools for reproducible science | SQL | Git | R | LaTeX | Markdown | Makefile

2020

**Software and Data Carpentry workshop**

Bios$^2$ training for graduate students — Online

The Unix Shell Data Management with SQL for Ecologists

2019 | 2018

**QCBS R Workshops**

Université de Sherbrooke & Université du Québec à Rimouski — Sherbrooke & Rimouski, CA

Instructor to graduate students Workshops: Introduction to R | Data management | Data Visualization | Control flow | Linear models

2019

**Probability and statistics**

GCI145 - Faculté de génie, Université de Sherbrooke — Sherbrooke, CA

Teaching assistant and lectured for undergraduate students Probability theory | Distribution | Descriptive statistics | Hypothesis testing | Linear models

2018

**Modelisation de la dynamique forestier du Quebec face au changement climatique**

Département de biologie, Université de Sherbrooke — Sherbrooke, CA

Guest lecturer in Plant Ecology (Presentation) to undergraduate students Covered modelling approaches in ecology | model construction and applications

2018

**Introduction to scientific programming**

BIO109 - Département de biologie, Université de Sherbrooke — Sherbrooke, CA

Teaching assistant for undergraduate students Pogramming concepts | Data management | Control flow | R language

## TRAINING

2022

**Data visualization and biodiversity modelling**

Summer school - Bios² — Magog, CA

Business intelligence | data user profiles | Conceptualize interactive dashboard | Figma

2020

**Mathematical Modelling in Ecology and Evolution**

Workshop - Bios² — Remote

Developing equations | Finding equilibria | analyzing stability | Maxima

2019

**Stage-based demographic models in ecology, evolution and conservation biology**

NERC Advanced Training - University of Oxford

Oxford, UK

Theory and practice | Matrix population models | Integral Projection Models

2019 **Biodiversity Modelling**
Summer school - Université de Sherbrooke
Magog, CA
Theory and practice on different biodiversity modelling techniques Statistics | Differential equations | Stochastic simulations | Climate change models | uncertainty tracking

2017 **Data-driven ecological synthesis**
DDES summer school - Université de Montréal
Montréal, CA
Good practices in scientific computing Data management | Mathematical tools | Data analysis tools | Parallel computing | Rreproducible science

2017 **Bayesian Statistics for Ecologists**
Summer school - Université de Sherbrooke
Magog, CA
Theory and practice using Stan | Probability theory | Likelihood | MCMC | Hierarchical modelling | Model comparison

## PUBLICATIONS

2026 **Unifying individual and metapopulation scales with stochastic population models: the effect of climate and competition on tree range limits**
In production. Preprint hosted on
Vieira, W., Gravel, D.

2026 **Sensitivity of tree species demography to climate and competition across their range**
Under review. Preprint hosted on
Vieira, W., MacDonald, A., Gravel, D.

2024 **Paying colonization credit with forest management could accelerate the range shift of temperate trees under climate change**
Ecological Modelling
• Vieira, W., Boulangeat, I., Brice, M., Bradley, R.L., Gravel, D.

2020 **Moderate disturbances accelerate forest transition dynamics under climate change in the temperate–boreal ecotone of eastern North America**
Global Change Biology
• Brice, M.H., Vissault, S., Vieira, W., Gravel, D., Legendre, P., Fortin, M.J.

2020 **Genetic and demographic aspects of Varronia curassavica Jacq. in a heterogeneous coastal ecosystem**
Anais da Academia Brasileira de Ciências
• Hoeltgebaum, M.P., Lauterjung, M.B., Montagna, T., Candido-ribeiro, R., Vieira, W., Bernardi, A.P., Cristofolini, C., Reis, M.S.D.